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Work, Power & Energy

Quick Revision Notes · Theory & Standard Results · JEE Main / Advanced · NEET

Work Done by a Constant Force

Definition

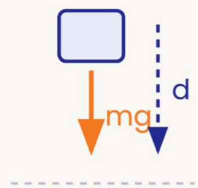
KEY

$$W = \vec{F} \cdot \vec{d} = Fd\cos\theta$$

- Scalar quantity · SI unit joule, $1\,\mathrm{J} = 1\,\mathrm{N{\cdot}m}$
- Frame-dependent — because displacement is itself frame-dependent (it differs between observers).

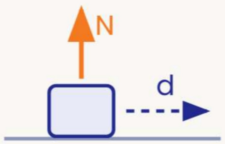
Trap: a large force can still do zero work — displacement and $\cos\theta$ decide, not the magnitude of F.

Sign of Work



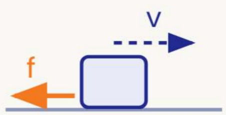
$\theta < 90^\circ \Rightarrow W > 0$

gravity on a falling block



$\theta = 90^\circ \Rightarrow W = 0$

normal force, block sliding



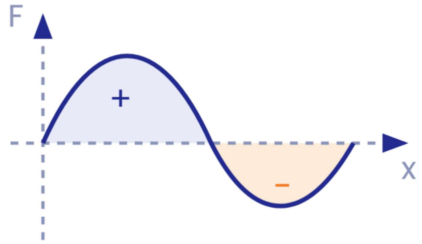
$\theta > 90^\circ \Rightarrow W < 0$

kinetic friction opposing sliding

Work by a Variable Force & Spring

Variable Force

$$W = \int \vec{F} \cdot d\vec{r} \;\; ; \qquad W_{1\text{-}D} = \int F\, dx$$



Work = signed area under the F–x graph; area below the axis counts negative.

Spring

KEY

$$W_{spring} = \frac{1}{2} k \left(x_1^2 - x_2^2 \right)$$

deformation changes $x_1 \to x_2$

$$W_{ext}\ (\text{quasi-static}) = \frac{1}{2} k x_2^2 - \frac{1}{2} k x_1^2$$

Trap: x in $\frac{1}{2}kx^2$ is the deformation from natural length — never the total length or the coordinate.

Kinetic Energy & Work–Energy Theorem

Kinetic Energy

$$K = \frac{1}{2}mv^2 = \frac{p^2}{2m} \quad ; \qquad p = \sqrt{2mK}$$

- $K \geq 0$ always; frame-dependent
- Same $p \Rightarrow$ lighter body has more K
- Same $K \Rightarrow$ heavier body has more p

Work–Energy Theorem

KEY

$$W_{net} = \Delta K = K_f - K_i$$

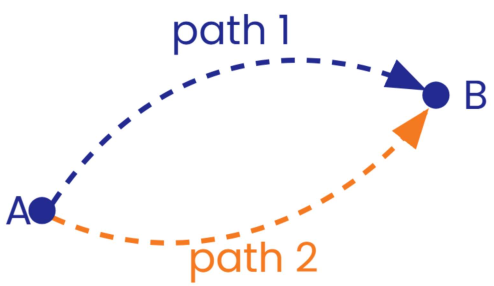
counts the work of **every** force — conservative, non-conservative, internal, external

Note: in a non-inertial frame the theorem still holds if the work done by the pseudo force is included.

Conservative Forces & Potential Energy

Conservative Force

- Work is path-independent
- Closed-loop work = 0
- Gravity, spring, electrostatic
- Non-conservative: friction, viscous drag



$W_{path\,1} = W_{path\,2}$

Potential Energy

KEY

$$\Delta U = -W_{cons}$$

defined only for conservative forces

$$U_{grav} = mgh \;\; ; \qquad U_{spring} = \frac{1}{2}kx^2$$

Reference level is arbitrary — only ΔU is physical.

Force from PE

$$F_x = -\frac{dU}{dx}$$

- Force points from high U toward low U
- Slope of the U–x graph gives $-F$

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Conservation of Mechanical Energy

Conservation

KEY

$$K + U = \text{constant}$$

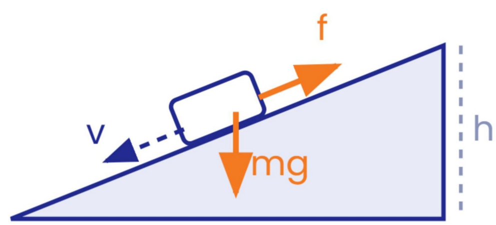
$$\Delta K + \Delta U = 0$$

valid if **only conservative** forces do work

With Non-Conservative Forces

$$W_{nc} = \Delta E = (K_f + U_f) - (K_i + U_i)$$

Heat by friction $= \mu N \times$ relative sliding distance



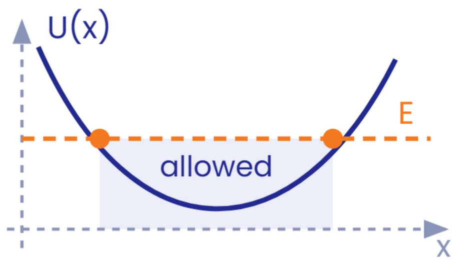
Energy Method

1. List all forces; classify conservative vs non-conservative
2. Choose a reference level for U
3. Write E_i, E_f using $\frac{1}{2}mv^2 + mgh + \frac{1}{2}kx^2$
4. Apply $W_{nc} = E_f - E_i$

U(x) Graphs & Equilibrium

Allowed Motion

$K = E - U \geq 0 \;\Rightarrow$ motion only where $U(x) \leq E$

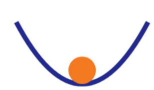


- Turning points where $U = E$ – there $v = 0$
- Region with $U > E$ is forbidden

Equilibrium

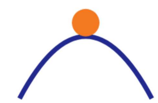
KEY

$$\frac{dU}{dx} = 0$$



Stable

$\frac{d^2U}{dx^2} > 0, U$ min



Unstable

$\frac{d^2U}{dx^2} < 0, U$ max

Neutral

U constant

Trap: force $= -$slope of $U(x)$ – it pushes toward lower U, not toward the origin.

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Power

Definitions

KEY

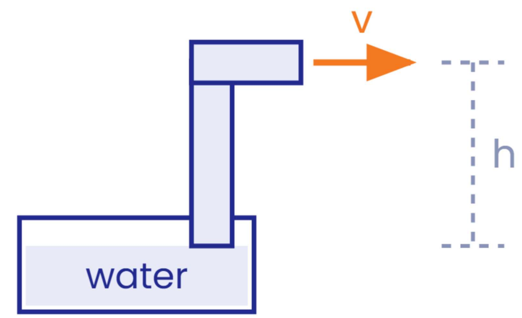
$$P_{avg} = \frac{W}{\Delta t} \quad ; \qquad P = \frac{dW}{dt} = \vec{F} \cdot \vec{v} = Fv\cos\theta$$

- SI unit: watt (W); $1\,\text{hp} = 746\,\text{W}$
- $1\,\text{kWh} = 3.6 \times 10^6\,\text{J}$ – a unit of **energy**, not power

Efficiency

$$\eta = \frac{P_{output}}{P_{input}}$$

Pump Raising Water



mass flow rate $\frac{m}{t}$, lifted through h, ejected at speed v

$$P = \frac{m}{t}\left(gh + \frac{v^2}{2}\right)$$

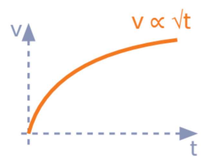
Motion at Constant Power

From Rest Under Constant P

KEY

$$mv\frac{dv}{dt} = P \;\Rightarrow\; v = \sqrt{\frac{2Pt}{m}} \;\Rightarrow\; v \propto t^{1/2}$$

$$x = \frac{2}{3}\sqrt{\frac{2P}{m}}\,t^{3/2} \;\Rightarrow\; x \propto t^{3/2}$$



Speed vs Position

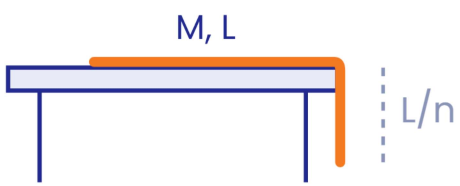
$$P = mv^2\frac{dv}{dx}$$

$$v = \left(\frac{3Px}{m}\right)^{1/3} \;\Rightarrow\; v \propto x^{1/3}$$

Note: constant power ≠ constant force: $F = P/v$ falls as speed builds. Top speed on a level road: $v_{max} = P/F_{resistive}$.

Standard Results: Chain & Spring–Block

Hanging Chain



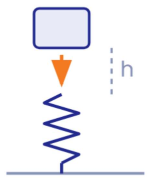
$\frac{1}{n}$th of the length hangs — work to pull it fully onto the table:

KEY

$$W = \frac{MgL}{2n^2}$$

hanging part's CoM rises by $\frac{L}{2n}$

Block Dropped on a Spring



$$mg\,(h + x_{max}) = \frac{1}{2}kx_{max}^2$$

- Released at natural length $(h = 0)$: $x_{max} = \frac{2mg}{k}$
- Gently lowered: $x_0 = \frac{mg}{k}$
- Speed is maximum at x_0

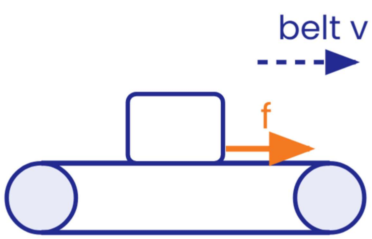
Trap: $v = 0$ at x_{max} but $a \neq 0$ there; $a = 0$ at $x_0 = mg/k$ where speed is maximum — never mix the two.

Work by Friction & Common Traps

MISC

Friction Can Do + Work

A block on an accelerating belt speeds up because friction on it acts **along** its motion.



Normal & Tension

- Normal force **does** work on a passenger in a moving lift — displacement is along N
- Ideal string tension does zero **net** work on the connected system
- A spring's internal work is non-zero

Traps Checklist

1. $K \geq 0$ — regions with $U > E$ are forbidden
2. $\Delta U = -W_{cons}$: mind the sign
3. Mechanical energy conserved only when $W_{nc} = 0$
4. In accelerating frames, include pseudo-force work
5. x in $\frac{1}{2}kx^2$ is deformation, not length

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